

Glacier Retreat in the Alps

Alpine glaciers store roughly seventy percent of the region's freshwater reserve. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Between 2000 and 2020 the average glacier front retreated by nearly one kilometre. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Meltwater timing shifts are already measurable in the Rhone and Po basins. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Downstream hydropower operators have re-licensed reservoirs to buffer the change. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Model ensembles project that half of today's glacier volume disappears by 2060. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

The remaining ice concentrates above four thousand metres of elevation. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Summer river discharge could drop by a quarter in dry years after mid-century. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.

Adaptation plans therefore prioritise storage, not just emission scenarios. Field surveys, satellite altimetry, and long-running mass-balance series all agree on the direction of the trend, differing only in the pace attributed to individual valleys.